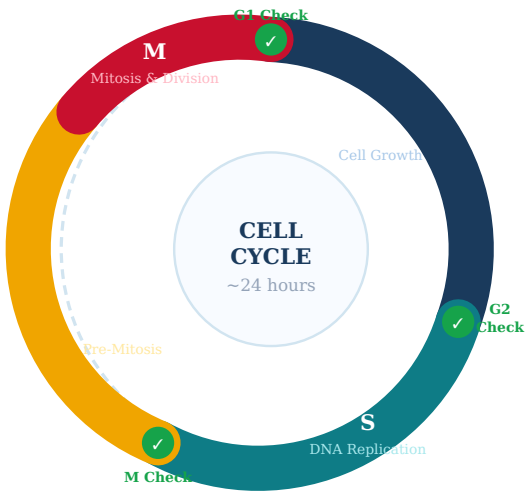
SECTION 1 OF 5 · LESSON 2

The Normal Cell Cycle

Before we understand what goes wrong, we need to understand what's supposed to happen.

Every cell in your body follows a highly regulated sequence of events before it divides. This sequence — the **cell cycle** — ensures that each new cell gets an exact, undamaged copy of the DNA.



The cell cycle — green checkmarks indicate quality-control checkpoints

THE FOUR PHASES

- **G1** — Cell grows, prepares for DNA copying
- **S** — DNA is replicated (copied)
- **G2** — Cell checks its work, prepares to divide
- **M** — Mitosis: cell splits into two identical daughter cells

WHY THIS MATTERS

A typical human cell divides roughly **50–70 times** in its lifetime. Each division must produce a perfect copy. Small errors in this process — especially if the cell doesn't catch them — can have serious consequences.

CHECK YOUR UNDERSTANDING

During which phase of the cell cycle does DNA replication occur?

A G1 phase — the cell grows and prepares

B S phase — synthesis of a new DNA copy

C G2 phase — the cell double-checks its DNA

D M phase — mitosis is happening

SECTION 2 OF 5

Checkpoints: The Cell's Rules

How does a cell "know" when it's safe to move forward — and when something is wrong?

ANALOGY TO BUILD YOUR MENTAL MODEL

Think of cell cycle checkpoints like **quality-control stations** on a factory assembly line. Before a car moves to the next stage of production, an inspector checks: Is this component correct? Is anything damaged? Only if everything passes inspection does the process move forward. The cell uses the same logic — with proteins instead of inspectors.

The Three Major Checkpoints

CHECKPOINT	WHEN IT OCCURS	WHAT IT CHECKS	KEY PROTEIN
G1 Checkpoint	End of G1 phase	Is the cell large enough? Is DNA undamaged? Are nutrients available?	p53, Rb protein
G2 Checkpoint	End of G2 phase	Was all DNA replicated correctly? Are there any double-strand breaks?	ATM/ATR kinase
M Checkpoint	During mitosis	Are all chromosomes properly attached to spindle fibers?	MAD2 protein

Key Term — Tumor Suppressor Gene: A gene that codes for a protein that slows or stops cell division. Think of it as the cell's "brake pedal." The most studied example is *TP53*, which codes for the p53 protein. p53 acts as the guardian of the G1 checkpoint — if DNA is damaged, p53 halts the cell cycle and triggers DNA repair.

PROTO-ONCOGENES

Normal genes that promote cell growth. When mutated, they become **oncogenes** — like a stuck accelerator. The cell keeps getting "go" signals even when it shouldn't divide.

TUMOR SUPPRESSOR GENES

Genes that code for "brakes." Both copies must be mutated to lose function (two-hit hypothesis). When both copies are broken, the cell loses the ability to slow itself down.

CHECK YOUR UNDERSTANDING

Which of the following best describes the role of the p53 protein?

A. It speeds up DNA replication in the S phase

B. It signals the cell to begin mitosis immediately

C. It halts the cell cycle if DNA damage is detected, allowing time for repair

D. It separates chromosomes during cell division

When Checkpoints Fail

What happens when the cell's quality-control system gets a mutation?

The core idea: Checkpoints only work if the proteins that run them are functioning correctly. If a gene that codes for a checkpoint protein mutates, the checkpoint can be bypassed — and a damaged or improperly copied cell moves forward anyway.

Two Ways Mutations Drive Cancer

GAIN-OF-FUNCTION MUTATION

Proto-oncogene → Oncogene

One copy mutated is enough. The protein becomes permanently "on," sending constant "divide now" signals. Like a gas pedal stuck to the floor.

Example: RAS oncogene — found mutated in ~30% of all cancers

LOSS-OF-FUNCTION MUTATION

Tumor Suppressor → Inactivated

Both copies must be mutated (Knudson's two-hit hypothesis). Once both copies are lost, the "brake" is gone and cell division is unchecked.

Example: TP53 mutation — found in over 50% of all cancer types

A CAR ANALOGY

Imagine a car with a **stuck gas pedal (oncogene)** and **no working brakes (tumor suppressor)**. Separately, each is dangerous. Together, they create a vehicle that accelerates with no way to stop — an almost perfect description of a cancer cell.

Why does it take more than one mutation? The cell has multiple, redundant safety systems. Most single mutations are caught by another backup checkpoint. Cancer usually requires several mutations, accumulated over time — which is why cancer risk increases with age.

St. Jude Connection: Researchers at St. Jude have extensively studied TP53 mutations in pediatric cancers. In some childhood cancers, TP53 mutations are inherited — which means some families carry elevated cancer risk. St. Jude's research has helped develop screening protocols to identify these families earlier.

CHECK YOUR UNDERSTANDING

A student says: "If someone has one mutated copy of a tumor suppressor gene, they will definitely get cancer." Is this correct? Why or why not?

- A Yes — one mutated copy is enough to completely lose checkpoint function
- B No — the second functional copy can still produce the checkpoint protein; both copies usually need to be mutated (two-hit hypothesis)
- C No — tumor suppressor genes never cause cancer, only oncogenes do

D Yes — the mutated copy will always override the normal copy

SECTION 4 OF 5

Cancer at the Cellular Level

Connecting what we've learned to what cancer actually is — and why St. Jude exists.

> Cancer is not one disease. It is **hundreds of diseases** — each caused by a unique combination of mutations in the cell cycle machinery of a specific cell type. What they share is uncontrolled cell division.

NORMAL CELL

- Checkpoints fully functional
- Divides when needed
- Undergoes apoptosis when damaged
- Stays where it belongs

PRE-CANCEROUS CELL

- 1–2 checkpoint mutations
- Backup systems still partially work
- Not yet uncontrolled
- Detectable by screening

CANCER CELL

- Multiple mutations accumulated
- Checkpoint systems bypassed
- Divides uncontrollably
- Can invade other tissues (metastasis)

> **What is Acute Lymphoblastic Leukemia (ALL)?**
>
> ALL is the most common childhood cancer — and one of the most studied at St. Jude. In ALL, lymphoblasts (immature white blood cells) undergo uncontrolled division in the bone marrow. Key mutations often involve tumor suppressor genes that normally regulate lymphocyte development. Because ALL affects cells that fight infection, patients experience immune system failure, anemia, and bone pain. **Today, St. Jude research has helped bring the survival rate for pediatric ALL to over 90%** — up from less than 10% in the 1960s.

Key Vocabulary Review

Cell Cycle Checkpoint
A quality-control pause point that verifies conditions are right before the cell advances to the next phase

Proto-Oncogene
A normal gene that promotes growth; becomes an oncogene when mutated, sending constant "divide" signals

Tumor Suppressor Gene
A gene that codes for "brake" proteins; when both copies are inactivated, the cell loses growth control

Two-Hit Hypothesis

The idea that both copies of a tumor suppressor gene must be mutated to fully lose its protective function

Apoptosis

Programmed cell death — a normal process where damaged cells destroy themselves; often bypassed in cancer cells

Metastasis

The spread of cancer cells from their original location to other tissues or organs via the bloodstream or lymph system

CHECK YOUR UNDERSTANDING

Which statement best explains why cancer is described as "many diseases" rather than one?

A Because it only occurs in certain organs

B Because it is caused by bacteria that vary by patient

C Because each cancer involves a unique combination of mutations in the cell cycle machinery of a specific cell type

D Because all cancers require the same treatment

Exit Ticket — Before Class Tomorrow

Your teacher will review these responses to shape tomorrow's discussion. Answer thoughtfully.

🎯 Complete before your next class session

These responses help your teacher know where to focus tomorrow's discussion. There are no trick questions — write what you actually think.

MULTIPLE CHOICE · QUESTION 1 OF 2

A cell's G1 checkpoint detects significant DNA damage. Under normal conditions, what is the most likely outcome?

- A The cell ignores the damage and proceeds to the S phase immediately
- B p53 is activated, the cell cycle pauses, and DNA repair mechanisms are triggered — or the cell undergoes apoptosis
- C The cell skips G1 and enters the M phase directly
- D The G2 checkpoint is notified but G1 continues normally

MULTIPLE CHOICE · QUESTION 2 OF 2

Which combination of mutations would be most likely to result in uncontrolled cell division?

- A A mutation that slows DNA replication slightly during S phase
- B A single mutated copy of a tumor suppressor gene
- C A mutation in a non-coding region with no effect on protein production
- D An activated oncogene (always-on growth signal) AND loss of both copies of a tumor suppressor gene

SHORT ANSWER · QUESTION 3 OF 3

Name one gene or protein that acts as a cell cycle checkpoint regulator. If this gene were to mutate and stop working, predict what might happen to the cell. Use evidence from what you learned in this module.

Write your response here. Think about: Which gene? What does it normally do? What happens if it's broken?

0 / minimum 40 characters recommended

Submit Exit Ticket →